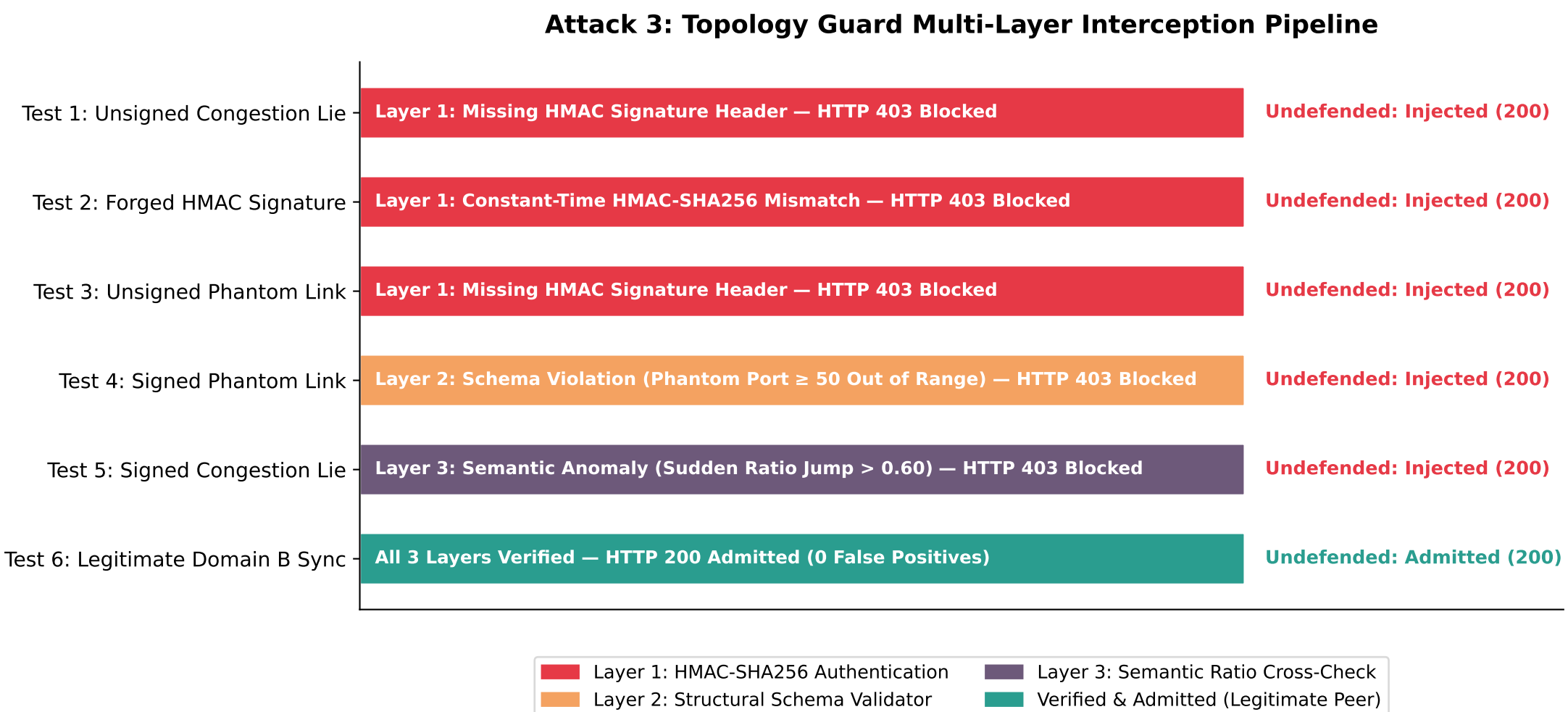
Attack 3: Topology Guard Multi-Layer Interception Pipeline
Test 1: Unsigned Congestion Lie
Layer 1: Missing HMAC Signature Header — HTTP 403 Blocked
Undefended: Injected (200)
Test 2: Forged HMAC Signature
Layer 1: Constant-Time HMAC-SHA256 Mismatch — HTTP 403 Blocked
Undefended: Injected (200)
Test 3: Unsigned Phantom Link
Layer 1: Missing HMAC Signature Header — HTTP 403 Blocked
Undefended: Injected (200)
Test 4: Signed Phantom Link
Layer 2: Schema Violation (Phantom Port ≥ 50 Out of Range) — HTTP 403 Blocked
Undefended: Injected (200)
Test 5: Signed Congestion Lie
Layer 3: Semantic Anomaly (Sudden Ratio Jump > 0.60) — HTTP 403 Blocked
Undefended: Injected (200)
Test 6: Legitimate Domain B Sync
All 3 Layers Verified — HTTP 200 Admitted (0 False Positives)
Undefended: Admitted (200)
Layer 1: HMAC-SHA256 Authentication
Layer 2: Structural Schema Validator
Layer 3: Semantic Ratio Cross-Check
Verified & Admitted (Legitimate Peer)